

A Shift-Invariant Statistical Measure Whose Filter Is a Countable Intersection of Ultrafilters

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Source Question and Status

Kadets and Seliutin ask in Problem 4.6 of their preprint whether a free filter generated by a shift-invariant statistical measure can equal the intersection of a countable family of free ultrafilters [1, p. 15].

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Problem 4.6. May a free filter $\mathfrak{F}$ generated by a shift-invariant statistical measure be equal to the intersection of some countable family of free ultrafilters?

Some finer than shift-invariance properties of statistical measures were discussed in [9], where the word “diffuse” instead of “statistical” was used. For filters generated by the corresponding measures, the respective variants of Problem 4.6 make sense as well.

The last questions concern the existence of minimal representations.

The substantially rebuilt published article, now with Tryba, explicitly answers several other preprint questions but retains this exact question as Problem 4.2 [2, Sec. 4.2]. This packet gives an affirmative answer in ZFC. More strongly, the countable family can be the full shift orbit of one ultrafilter. Status: *candidate full solution, likely valid*, subject to human review.

Stone–Čech Setup

Put $\mathbb{N} = \{1, 2, \dots\}$ and $\mathbb{N}^* = \beta\mathbb{N} \setminus \mathbb{N}$. Its points are the free ultrafilters on $\mathbb{N}$. For $A \subset \mathbb{N}$, write

$$A^* = \{p \in \mathbb{N}^* : A \in p\}.$$

These sets are clopen, $(A \cup B)^* = A^* \cup B^*$, and they are disjoint when A and B are disjoint.

Let $s(n) = n + 1$. The extension βs restricts to a homeomorphism $T : \mathbb{N}^* \rightarrow \mathbb{N}^*$. Indeed, the extension of a predecessor map is its inverse on $\mathbb{N}^*$: the two compositions differ from the identity on at most one natural number, and maps which agree off a finite set have the same extension on free ultrafilters. With the convention of the source,

$$T(p) = p + 1, \quad T(A^*) = (A + 1)^*. \quad (1)$$

Main Result

Theorem 1. *There are a shift-invariant statistical measure μ on $\mathbb{N}$ and a countably infinite family of pairwise distinct free ultrafilters $\{\mathfrak{U} + n : n \in \mathbb{Z}\}$ such that*

$$\mathfrak{F}_\mu = \{A \subset \mathbb{N} : \mu(A) = 1\} = \bigcap_{n \in \mathbb{Z}} (\mathfrak{U} + n).$$

Consequently, preprint Problem 4.6 and published Problem 4.2 have an affirmative answer.

Proof. Choose a nonempty minimal closed T -invariant subset K of $\mathbb{N}^*$. Such a set exists: start with the closure of any two-sided orbit and apply Zorn's lemma to the nonempty closed invariant subsets; intersections of descending chains remain nonempty by compactness. Fix $\mathfrak{U} \in K$. Minimality gives

$$K = \overline{\{T^n \mathfrak{U} : n \in \mathbb{Z}\}}. \quad (2)$$

For $N \geq 1$, let

$$\nu_N = \frac{1}{N} \sum_{j=0}^{N-1} \delta_{T^j \mathfrak{U}}.$$

The regular Borel probability measures on K form a weak-* compact subset of $C(K)^*$, so (ν_N) has a weak-* convergent subnet; call its limit ν . For every $f \in C(K)$,

$$\int f \circ T d\nu_N - \int f d\nu_N = \frac{f(T^N \mathfrak{U}) - f(\mathfrak{U})}{N} \rightarrow 0.$$

Thus ν is T -invariant.

Its support is a nonempty closed T -invariant subset of K . Nonemptiness does not require metrizability: if every point had a null open neighborhood, compactness would give a finite null cover of K , contrary to $\nu(K) = 1$. Minimality therefore gives

$$\text{supp } \nu = K. \quad (3)$$

Define, for every $A \subset \mathbb{N}$,

$$\mu(A) = \nu(A^* \cap K). \quad (4)$$

The clopen identities above show finite additivity and nonnegativity, and $\mu(\mathbb{N}) = 1$. If A is finite then $A^* = \emptyset$, so in particular $\mu(\{k\}) = 0$ for every k . Hence μ is a statistical measure. Equation (1), invariance of K , and invariance of ν give

$$\mu(A + 1) = \nu(T(A^* \cap K)) = \nu(A^* \cap K) = \mu(A),$$

so it is shift-invariant.

Full support is exactly what identifies its measure-one filter. Namely,

$$\mu(A) = 1 \iff K \subset A^*. \quad (5)$$

The forward implication holds because, if $K \setminus A^*$ were nonempty, it would be a nonempty relatively open subset of the full-support space K and hence would have positive ν -measure. The reverse implication is immediate. Since A^* is closed, (2) and (5) now give

$$\begin{aligned} A \in \mathfrak{F}_\mu &\iff K \subset A^* \\ &\iff T^n \mathfrak{U} \in A^* \quad (n \in \mathbb{Z}) \\ &\iff A \in \mathfrak{U} + n \quad (n \in \mathbb{Z}). \end{aligned}$$

This proves the asserted equality. Every $\mathfrak{U} + n$ is free, and their intersection contains the Fréchet filter, so $\mathfrak{F}_\mu$ is free.

It remains only to check that the displayed family has no repetitions, as the source uses “family” to mean a set. Suppose $T^m \mathfrak{U} = \mathfrak{U}$ for some $m \geq 1$. Let

$$E = \bigcup_{k \geq 0} \{2km + 1, \dots, (2k + 1)m\}.$$

Then $E - m := \{r : r + m \in E\} = \mathbb{N} \setminus E$. But $T^m \mathfrak{U} = \mathfrak{U}$ would imply $E \in \mathfrak{U}$ if and only if $E - m \in \mathfrak{U}$, contradicting the ultrafilter dichotomy. Thus the shifts are pairwise distinct. $\square$

Why the Construction Works

A naive positive weighted sum of the orbit ultrafilters has exactly the desired measure-one filter, but no positive probability weights on an infinite orbit are translation invariant. Passing from the orbit to its compact closure repairs invariance; passing to a minimal orbit closure forces every nonempty relative open set to have positive measure. That full-support property preserves the measure-one filter exactly, so no ultrafilters are lost in the compactification step.

The same proof gives a useful criterion: whenever the orbit closure of a free ultrafilter carries a full-support invariant probability, its two-sided shift intersection is generated by a shift-invariant statistical measure. Minimal points guarantee the criterion in ZFC.

Scope, Novelty Search, and Verification

This is an existence result, exactly as the word “May” in the source asks. It does not assert that every shift-invariant statistical-measure filter has a countable representation; the source already records examples showing otherwise.

Cheap run indexes contained no answer. A bounded search through 17 August 2026 used the arXiv id, exact question, published title and DOI, and combinations of *shift-invariant statistical measure*, *countable ultrafilter intersection*, *minimal dynamics*, and $\mathbb{N}^*$. It found the source and the 2022 article retaining the problem, but no later answer. This is not a priority claim.

The proof was audited for the source’s shift orientation, finite additivity, vanishing on finite sets, weak-* compactness without metrizability, full support, freeness, and distinctness of the orbit ultrafilters. No numerical verification is relevant.

Human Review Recommendation

Check equation (1), the existence of the invariant weak-* subnet, and the equivalence (5). Those are the three structural passage points; the remaining argument is direct Stone duality.

References

- [1] V. Kadets and D. Seliutin, *Conglomerated filters, statistical measures, and representations by ultrafilters*, arXiv:2012.02866 (2020).
- [2] V. Kadets, D. Seliutin, and J. Tryba, *Conglomerated filters and statistical measures*, J. Math. Anal. Appl. **509** (2022), 125955, doi:10.1016/j.jmaa.2021.125955.